John Boerchers

AEROSPACE & PROPULSION · COMPUTATIONAL FLUID DYNAMICS · TURBULENT REACTING FLOWS

973-479-0494 · johnnyboerchers@gmail.com · linkedin.com/in/john-boerchers · johnboerchers.github.io ·

SUMMARY

Fifth-year PhD candidate at Princeton (exp. May 2027) developing high-fidelity combustion models for hypersonic and supersonic propulsion. Expertise in manifold-based LES turbulence-chemistry interaction, high-speed reacting flow physics, and scientific software development (C++, Fortran, Python). Proven experience bridging fundamental research with propulsion-relevant applications, including GPU-accelerated combustion codes and collaboration with national laboratories.

TECHNICAL SKILLS

Languages: Python (proficient), Fortran (intermediate), C++ (intermediate)

HPC & Parallel: MPI / mpi4py, OpenMP, OpenACC (GPU)

Scientific Tools: NumPy, SciPy, MATLAB, Git, Bash / Linux, LaTeX

Machine Learning: PyTorch

Math Foundations: PDEs / ODEs, linear algebra, numerical methods & algorithms, optimization

EDUCATION

Princeton University — 2022 – May 2027 (expected)

Ph.D. Candidate, Mechanical & Aerospace Engineering

M.A. in Mechanical & Aerospace Engineering (conferred 2024) · Graduate Certificate in Computational Science & Engineering (in progress) · Advisor: Prof. Michael E. Mueller · MAE Department Fellowship (2023)

Pennsylvania State University — 2018 – 2022

B.S., Aerospace Engineering — Schreyer Honors College, Summa Cum Laude

GPA 3.99 · Honors Thesis: *A Viscous-Inviscid Channel Flow Approach for Preliminary Wind Tunnel Design* · Richard W. Leonhard Scholarship · Evan Pugh Scholar Award

RESEARCH EXPERIENCE

Computational Turbulent Reacting Flow Laboratory, Princeton University — 2022 – Present

Graduate Research Assistant — Advisor: Prof. Michael E. Mueller

- Developing manifold-based combustion models that extend LES capability to high-speed reacting flows, enabling efficient high-fidelity simulations for rotating detonation engine (RDE) and scramjet propulsion research.
- Implementing and validating models within PeleC, an AMR-enabled exascale reacting flow solver, targeting rotating detonation combustor configurations.
- Collaborated with NREL to benchmark manifold models against DNS data for supersonic combustion.

Lawrence Livermore National Laboratory — June – Aug 2022

Research Intern — Advisor: Boyan Lazarov

- Built a physics-constrained neural network (PyTorch) trained on DNS data to accelerate topology optimization of porous-media flow problems — reducing design iteration cost by leveraging data-driven surrogates.

Aircraft Propulsion Research Group, Penn State — May 2021 – May 2022

Undergraduate Researcher — Advisor: Prof. David K. Hall

- Developed a viscous-inviscid coupled numerical tool for preliminary wind tunnel aerodynamic design and performance analysis.

High Pressure Combustion Laboratory, Penn State — Nov 2019 – May 2021

Undergraduate Researcher — Advisor: Prof. Richard Yetter

- Designed a CO_2 laser ignition experiment to quantify energy input and flame propagation for energetic solid propellants; contributed to published study on piezoelectric effects in nano-aluminum composites.

SELECTED PROJECTS

Princeton GPU Hackathon — June 2023

- Partnered with NVIDIA mentor to GPU-accelerate combustion chemistry kernels using OpenACC in a thermochemical solver; optimized parallel execution and GPU memory management to achieve speedup on combustion-critical code paths.

TEACHING & MENTORSHIP

Teaching Assistant, Princeton University 2024 – 2026

- *MAE 221: Thermodynamics* — Fall 2026
- *MAE 426: Rocket & Air-Breathing Propulsion Technology* — Spring 2026
- *MAE 427: Energy Conversion & the Environment* — Spring 2025 · **E-Council Teaching Award 2026**
- *MAE 335: Compressible Flows* — Fall 2024 · **Crocco Teaching Award 2025**

Undergraduate Thesis Mentor 2024 – 2025

Mentored Laura T. Thompson on RDE numerical modeling (hydrogen/methane fuel composition). Thesis earned **MAE Best Undergraduate Thesis Award** and the **SEAS Thesis Award**.

PUBLICATIONS & PRESENTATIONS (SELECTED)

A Thermodynamically Consistent Manifold Model for Premixed Deflagrations & Detonations
J.B. Boerchers, M.X. Yao, L.T. Thompson, M.E. Mueller · Submitted to Combustion and Flame, 2026 · arXiv preprint

Implementation of a Thermodynamically Consistent Manifold Model for Premixed Detonations and Deflagration
J.B. Boerchers, L.T. Thompson, M.X. Yao, M.E. Mueller · APS Division of Fluid Dynamics, 2025

A Thermodynamically Consistent Manifold Model for Premixed Detonations and Deflagration
J.B. Boerchers et al. · International Conference on Numerical Combustion, 2025

Modeling Enthalpy Variations with Mixture Fraction in High-Speed Turbulent Combustion
J.B. Boerchers, M.E. Mueller · US National Combustion Meeting, 2025

Modeling and Topology Optimization of Flow Problems in Porous Media Using Neural Networks
M. Schmidt, J.B. Boerchers, B. Lazarov · ASME IMECE, 2022